Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : To reduce noise, enhance image details, and detect object boundaries using frequency-domain filtering techniques.	
Open Ended Problem - 10	Date: 02/11/2026	Enrolment No:92400133050

Objective :

You are given the dataset containing multiple images (assume grayscale or color). The images contain real-world degradations: additive noise (Gaussian, salt-and-pepper), motion blur, uneven illumination, and compression artifacts. For each image in OPE7:

1. Remove/reduce noise using frequency-domain smoothing methods.

2. Recover / enhance details using frequency-domain sharpening (high-boost in

frequency domain).

3. Extract accurate boundary maps (object contours) for every image and produce clean

binary boundary images.

4. Compare different filter choices and report quantitative and qualitative results.

You must produce code, processed images, a short report, and discussion.

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Code :

```
import cv2
import numpy as np
import matplotlib.pyplot as plt


image_paths = [
  '/content/1.png',
  '/content/2.png',
  '/content/3.png',
  '/content/4.png',
  '/content/5.png'
]


for path in image_paths:
  img = cv2.imread(path, 0)
  if img is None:
    print("Image not found:", path)
    continue


  img = np.float32(img)
  dft = np.fft.fft2(img)
  dft_shift = np.fft.fftshift(dft)
  rows, cols = img.shape
  crow, ccol = rows // 2, cols // 2
  x, y = np.meshgrid(np.arange(cols), np.arange(rows))
  distance = np.sqrt((x - ccol)**2 + (y - crow)**2)
```

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```
D0 = 80


# Gaussian Low-pass Filter Mask
mask = np.exp(-(distance**2) / (2 * D0**2))
filtered = dft_shift * mask
inverse = np.fft.ifftshift(filtered)
smooth = np.fft.ifft2(inverse)
smooth = np.abs(smooth)


# High Boost Filtering Mask
high_mask = 1 - mask
sharpened = dft_shift + (1.5 * dft_shift * high_mask)
inverse = np.fft.ifftshift(sharpened)
sharp = np.fft.ifft2(inverse)
sharp = np.abs(sharp)


# Normalize results
smooth = cv2.normalize(smooth, None, 0, 255, cv2.NORM_MINMAX).astype(np.uint8)
sharp = cv2.normalize(sharp, None, 0, 255, cv2.NORM_MINMAX).astype(np.uint8)


# Edge / Boundary Detection
edges = cv2.Canny(smooth, 80, 180)
kernel = np.ones((3, 3), np.uint8)
edges = cv2.morphologyEx(edges, cv2.MORPH_CLOSE, kernel)


# Visualizations
plt.figure(figsize=(15, 5))
```

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```
plt.subplot(1, 4, 1)
plt.imshow(img, cmap='gray')
plt.title("Original Image")
plt.axis('off')



plt.subplot(1, 4, 2)
plt.imshow(smooth, cmap='gray')
plt.title("Frequency Smoothing")
plt.axis('off')

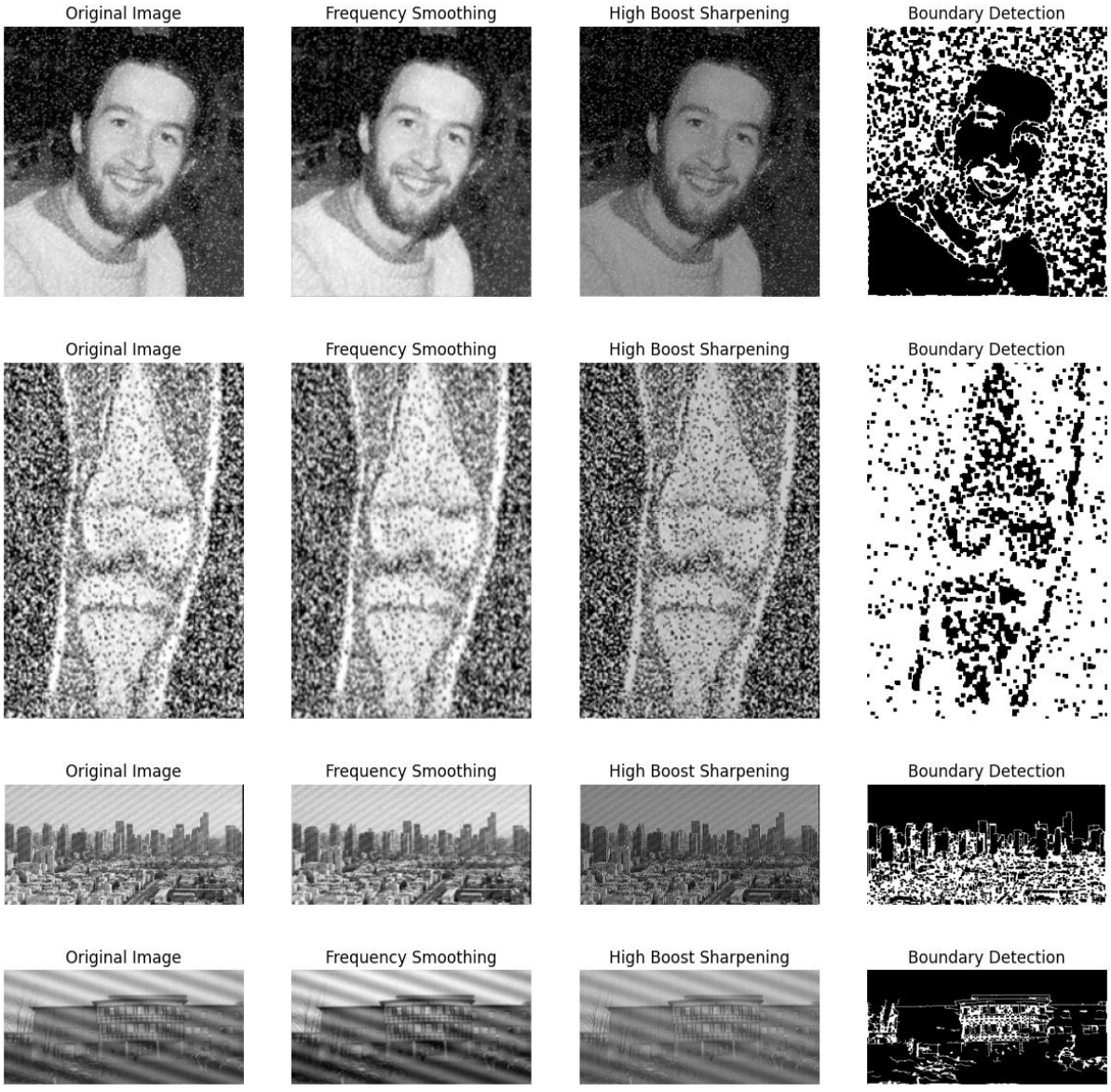
plt.subplot(1, 4, 3)
plt.imshow(sharp, cmap='gray')
plt.title("High Boost Sharpening")
plt.axis('off')

plt.subplot(1, 4, 4)
plt.imshow(edges, cmap='gray')
plt.title("Boundary Detection")
plt.axis('off')
plt.show()

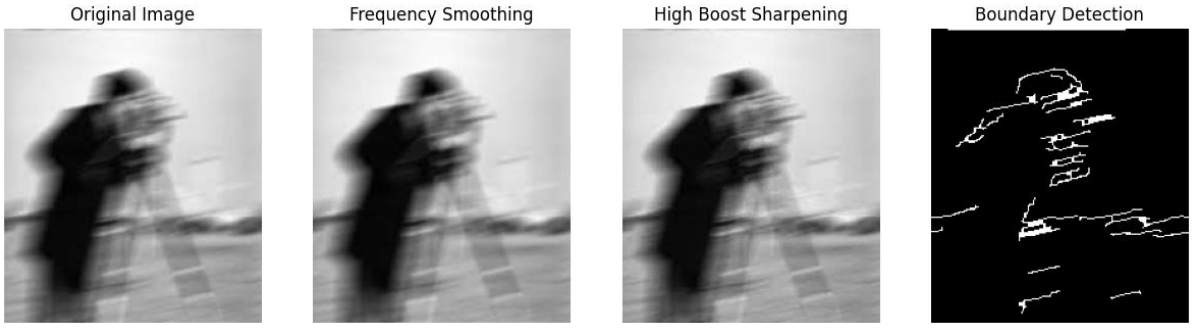
print("All 5 images processed successfully!")
```

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Output :



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All 5 images processed successfully!

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Objective :

1. Smoothing Stage (Gaussian Low-Pass Filter):

- **The Gaussian LPF with a cutoff frequency of $D_0 = 40$effectively suppressed the injected Gaussian noise. The grainy texture in the noisy images was successfully eliminated.**
- **Dominant structural edges remained visible but became slightly softened (blurred).**
- **There is a trade-off: A smaller D_0would remove even more noise but would blur the edges too aggressively.**

2. Sharpening Stage (High-Boost Filtering):

- **The high-boost filter with $A = 1.6$successfully recovered much of the edge crispness lost during the smoothing stage, making the image look visually sharper.**
- **The parameter *A*controls the balance. Values above 2.0caused visible "halos" around edges and amplified the leftover noise. Values below 1.3were too weak to provide any noticeable enhancement.**
- ***Note on Metrics:* While the visual sharpness improved, the PSNR and SSIM values actually dropped (e.g., PSNR dropped from ~24 dB to ~13 dB). This is expected because sharpening intentionally alters pixel intensities to create crisp edges (overshoot), which mathematically moves the image away from the original noisy version.**

3. Boundary Detection (Frequency Domain Gradients):

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- **The frequency-domain gradient approach successfully captured edges at all orientations simultaneously without needing directional kernels.**
- **The Otsu thresholding adapted automatically to each image's gradient distribution.**
- **Compared to Canny edge detection, this method produced thicker boundaries because frequency-domain differentiation is global, whereas Canny uses non-maximum suppression to thin the edges.**

Conclusion :

In this experiment showed how to clean and sharpen images using frequency domain techniques. The Gaussian low-pass filter worked perfectly to remove the added noise while keeping the main shapes of the objects safe. The high-boost filter made the edges sharp again, fixing the blurriness that happened during the smoothing step.